Phạm Thanh Lâm

Network Engineer · Computer Networks & Data Communications

UIT graduate with hands-on experience in network security, high-availability infrastructure, and AI security. Seeking a full-time role in network engineering or security operations.

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EXPERIENCE

FSI — Công ty Cổ phần Công nghệ FSI
Mar 2026 - Jun 2026

Network & Security Engineer Intern

26/22 Đường số 03, Cư Xá Đô Thành, Phường Bàn Cờ, TP. HCM

- Configured and tested **F-DDH BOX** appliance (C2 traffic simulation via ICMP/TCP, log & alert validation); supported live customer demos at Viettel Đắk Lắk and Tây Ninh Provincial Innovation Center.
- Assisted enterprise network deployments: fiber optic splice repair (Cisco L3, White Palace), **Catalyst 9200L** IOS 17.12 upgrade & StackWise study, **Peplink 2500EC SD-WAN** planning; on-site logistics at Viettel IDC.
- Investigated **VPN bypass techniques** (Tailscale, Cloudflare WARP) against NGFW/non-NGFW perimeters; compiled presale security & regulatory documentation for customer advisory.

PROJECTS

HA Firewall & AI-driven Security Monitoring — Thesis (KLTN)
Jan - May 2026

- Built **Active-Passive OPNsense cluster** (CARP/pfsync) with HAProxy SSL offloading and **ModSecurity WAF (OWASP CRS)** for defense-in-depth; measured failover time and pfsync overhead.
- Deployed **Wazuh SIEM** pipeline; generated custom attack dataset (DVWA + Kali Linux); trained **Random Forest / Logistic Regression** classifier targeting FP rate reduction from >30% → <15%.

`OPNsense` `CARP/pfsync` `HAProxy` `ModSecurity` `OWASP CRS` `Wazuh SIEM` `Random Forest` `Python` `VMware`

AI Shield Firewall — LLM Traffic Inspection Proxy — Personal Project
2025 - 2026

- Built a **5-layer AI API inspection proxy** on AWS (EC2 t3.small, DynamoDB, Docker) protecting enterprises from data leakage when integrating LLMs; layers cover Regex, Presidio NLP, **Prompt Injection (EN+VI)**, per-org Policy Engine, and Output Inspection.
- Achieved **100% detection rate** on 23-sample bilingual benchmark (BLOCK/REDACT/ALLOW), **21/21 unit tests passed**; proxy overhead <15ms when co-located; Dockerized with `restart: always` for HA.
- Implemented **per-org Policy Engine** with DynamoDB-backed keyword blacklist and message-length enforcement; **IAM Role** credential management (no hardcoded keys); live monitoring dashboard with auto-refresh audit log visualization.

`Python` `FastAPI` `AWS EC2` `DynamoDB` `Docker` `Presidio NLP` `Prompt Injection` `IAM Role` `API Gateway`

AutoCTI — Autonomous Cyber Threat Intelligence Agent — Personal Project
2025 - 2026

- Designed and deployed a **24/7 automated CTI pipeline** on AWS (EC2, SQS, API Gateway, DynamoDB) operating at ~$0/month; Dockerized with `restart: always` for high-availability on EC2.
- Integrated **LangGraph ReAct Agent** with Gemini 2.5 Flash and **Tavily real-time search** for autonomous threat analysis; replaced Playwright with Tavily, reducing container RAM by ~80%.
- Engineered production-grade reliability: **threading.Lock()** for SQS race-condition prevention, Exponential Backoff for 429/503 handling, **IAM Role** credential management, and DynamoDB persistent conversation memory.

`AWS EC2` `SQS` `DynamoDB` `API Gateway` `LangGraph` `Python` `Docker` `Gemini AI` `Telegram Bot API`

CERTIFICATIONS & LEARNING

Networking Basics
Cisco Networking Academy — In progress

Networking Devices and Initial Configuration
Cisco Networking Academy — In progress

Endpoint Security
Cisco Networking Academy — In progress

Network Defense
Cisco Networking Academy — In progress

EDUCATION

University of Information Technology (UIT – VNU-HCM)

B.Sc. Computer Networks & Data Communications

GPA 7.72 / 10 · 2022 - 2026

Relevant courses:
Network Performance Evaluation · DevOps · Programmable Networks · Systems Programming

SKILLS

NETWORKING
TCP/IP · Routing/Switching · High Availability · CARP/pfsync · SDN/OpenFlow · SD-WAN · Cisco IOS · StackWise

SECURITY
OPNsense · Wazuh SIEM · AI Security · ModSecurity WAF · Prompt Injection · HAProxy · Suricata IDS/IPS · SNMP v2c/v3

CLOUD
AWS · Docker · EC2 · DynamoDB · API Gateway · IAM · Azure

SYSTEMS
Linux · Docker · VMware · Windows Server

DEV & TOOLS
Python · FastAPI · Git · C++ · Shell Scripting · LangGraph · Presidio NLP · scikit-learn

LANGUAGES

Vietnamese — Native
English — Technical reading